मधेश २०८०
प्रदेश लोक सेवा आयोग, मधेश प्रदेश
जनकपुरधाम।
ईन्जिनियरिङ्ग सेवा, सिभिल समूह, जनरल, हाईवे, ईरिगेशन, स्यानिटरी उप-समूह, अधिकृत सातौ तह वा सो सरह पदको खुला प्रतियोगितात्मक लिखित परीक्षा
मिति: २०८०/०६/१३ | समय : ३ घण्टा | पत्र : द्वितीय | पूर्णाङ्क : १००
विषय : Technical Subject
निम्न प्रश्नहरुको उत्तर छुट्टाछुट्टै उत्तरपुस्तिकामा लेख्नुपर्नेछ अन्यथा उत्तरपुस्तिका रद्द हुनेछ।

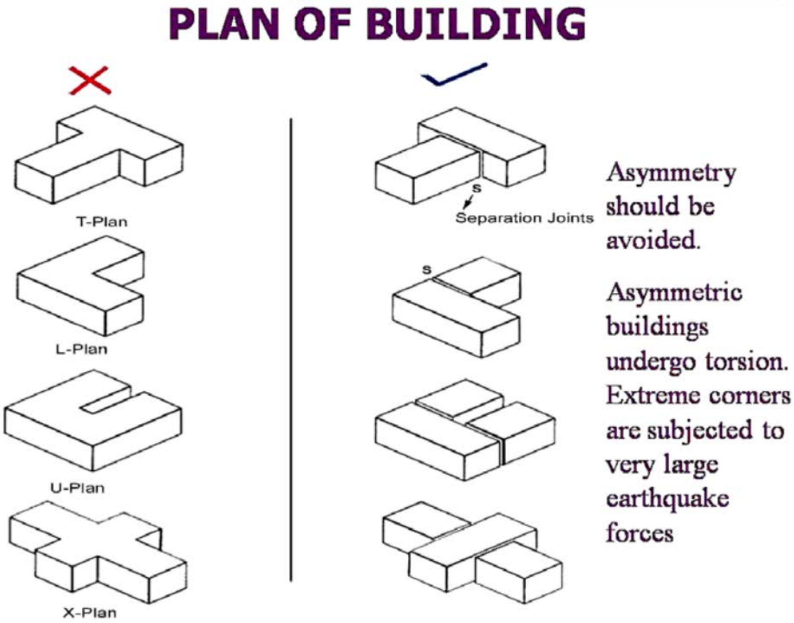
Section-A (25 Marks)

1. **As per Nepal National Building code, elaborate on characteristics and requirements of Earthquake resistant building. What are the pertinent factors to be considered in structural design? [10]**

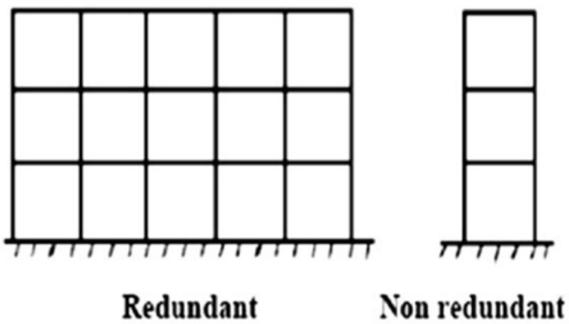
Answer:
Earthquake-resistant buildings are designed to prevent collapse during a major earthquake and minimize damage during minor ones. In Nepal, NBC 105:2020 provides the seismic design requirements.

Characteristics and Requirements (NBC 105):

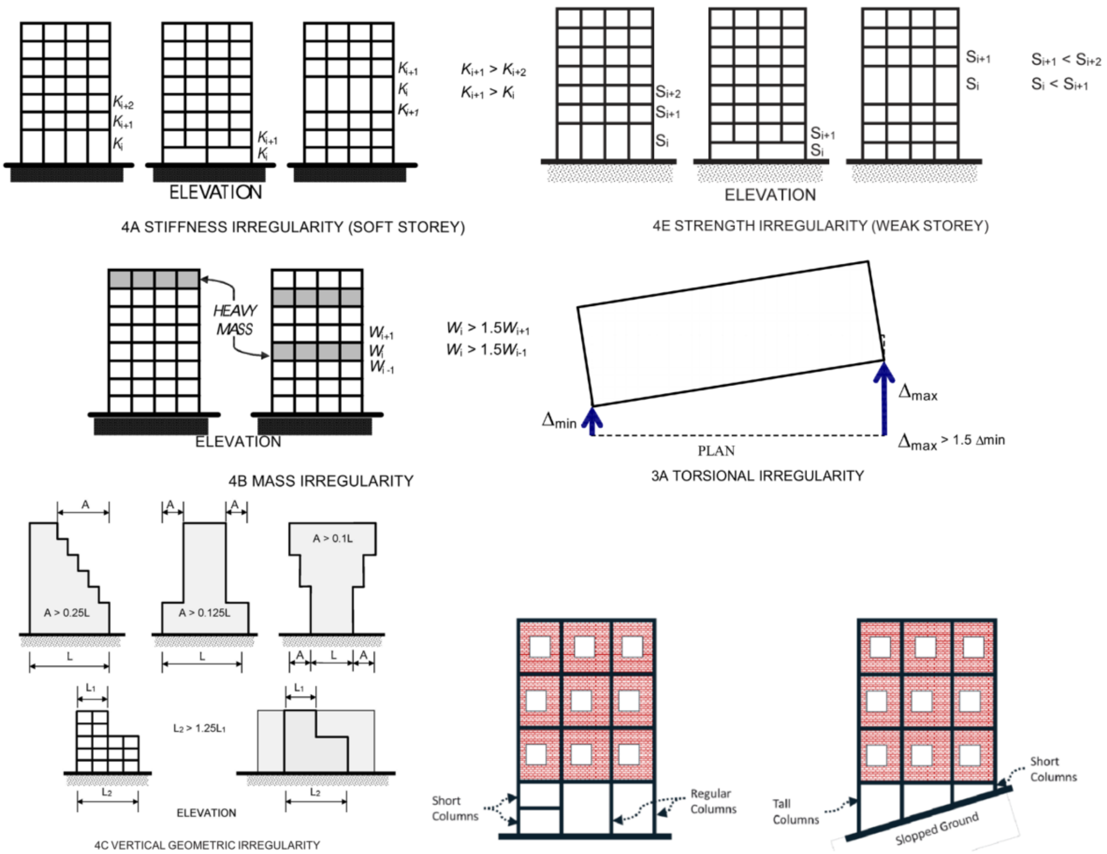
a) Symmetry and Regularity: Buildings should be symmetrical in plan and elevation to avoid torsional effects.



b) Ductility: The ability of a structure to undergo large deformations without collapsing. This is achieved through ductile detailing (e.g., proper stirrup spacing in beams/columns).
c) Redundancy: Multiple load paths should be available so that if one member fails, others can carry the load.



d) Integrity (Tying Together): All structural elements must be effectively tied together to act as a single unit (e.g., providing plinth beams and lintel bands).
e) Lightweight Construction: Seismic force ($F = m \cdot a$) is directly proportional to mass; hence, using lighter materials reduces the force.
f) Avoid Structural Irregularities: To ensure a building's safety, especially in seismic zones, it is critical to avoid or mitigate structural irregularities. These are typically classified into plan (horizontal) and vertical irregularities. Buildings with simple, symmetrical geometry and uniformly distributed mass and stiffness are significantly less vulnerable to structural failure.



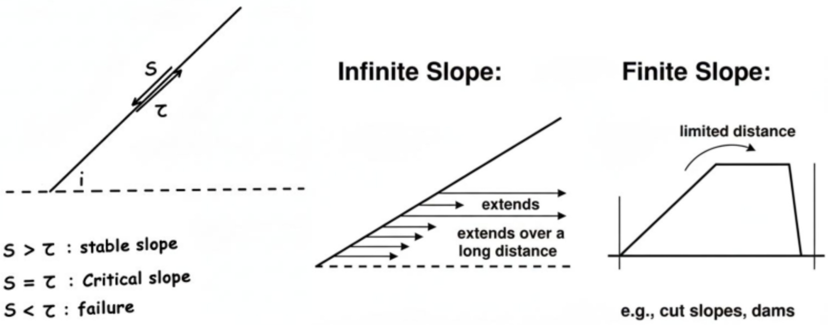
Pertinent Factors in Structural Design:

a) Seismic Hazard Factor (Z): Depends on the location in Nepal (e.g., Kathmandu has a higher Z than some Terai regions).
b) Importance Factor (I): Higher for essential buildings like hospitals ($I = 1.5$) compared to residential ($I = 1.0$).
c) Soil Classification: Type A (Hard Rock), Type B (Medium Soil), or Type C (Soft Soil). Soft soil amplifies ground motion.
d) Ductility Factor (R): Depends on the structural system (e.g., Moment Resisting Frame vs. Shear Wall).
e) Load Combinations: Considering $DL + LL \pm EQL$ as per code standards.
f) Fundamental Period (T): The natural time period of the building used to determine the design response spectrum.

2. Discuss the various causes of slope movement and failure. Explain different ways of stabilizing slopes. [$4 + 6 = 10$]

Answer:

Slope failure occurs when the shear stress exceeds the shear strength of the soil/rock mass.



Causes of Slope Failure:

A. Factors Increasing Shear Stress (τ)

Shear stress represents the "driving forces" pulling material downslope.

- **Increase in Unit Weight:** Saturation from heavy rainfall or snowmelt replaces air in pore spaces with water, which is heavier. This added mass increases the gravitational force pulling material down.
- **External Loading:** Surcharges from new construction, vehicles, or the dumping of mining waste add weight to the upper portion of a slope.
- **Seepage Forces:** Water flowing through soil mass exerts a "drag force" in the direction of flow, adding to the driving stresses.

- **Steepening/Erosion:** Removing lateral or underlying support through toe erosion (by rivers or waves) or human excavation increases the slope's inclination, thereby increasing the tangential component of gravity.
- **Dynamic Loading:** Earthquakes, blasting, or even heavy traffic generate transient stresses and vibrations that momentarily increase the driving forces and can trigger sudden movement.

B. Factors Decreasing Shear Strength (s)

Shear strength is the internal resistance of the soil or rock to sliding.

- **Increase in Pore Water Pressure (u):** High water pressure within soil pores reduces the **effective stress** ($\sigma' = \sigma - u$), which directly lowers the frictional resistance between particles.
- **Erosion and Weathering:** Physical and chemical weathering (e.g., freeze-thaw cycles or dissolution of cementing minerals) breaks down rock joints and soil cohesion.
- **Moisture Changes in Clay:** Clays, especially "expansive" types like smectite, swell when wet, pushing particles apart and losing nearly all strength.
- **Disturbance/Liquefaction:** Cyclic loading or shocks can cause loose, saturated soils to lose grain-to-grain contact, causing them to behave like a liquid—a process known as **liquefaction**.
- **Vegetation Removal:** Roots act as a natural "anchor," and their removal (by fire or deforestation) eliminates this reinforcement and increases water infiltration.

Stabilizing Ways:

Stabilization focuses on either reducing driving forces or increasing resisting forces.

- **Geometric Techniques:**
  - **Flattening:** Reducing the overall slope angle to lower shear stress.
  - **Benching:** Creating steps to interrupt failure surfaces and manage drainage.
  - **Buttressing:** Adding heavy material (like rock berms) to the toe to provide a counter-balancing weight.
- **Hydrological Techniques (Drainage):**
  - **Surface Drainage:** Using ditches and interceptor drains to divert water before it infiltrates.
  - **Subsurface Drainage:** Installing horizontal drains or vertical wells to lower the groundwater table and reduce pore water pressure.
- **Mechanical & Structural Reinforcement:**
  - **Soil Nailing & Rock Bolting:** Inserting steel rods or anchors to "tie" unstable material back to stable ground.
  - **Retaining Structures:** Building concrete walls, gabion baskets (wire cages filled with rock), or using geosynthetics to provide structural support.
  - **Shotcrete:** Spraying a layer of concrete over the slope face to prevent weathering and surface raveling.
- **Bioengineering:**
  - **Vegetation:** Planting deep-rooted shrubs and grasses to bind soil together and naturally remove moisture via transpiration.

3. What is bearing capacity of soil? What are the factors that affect the bearing capacity? What are the different design types that are usually selected for shallow as well as deep foundations? [5]

Answer:

Bearing Capacity (BC) is the maximum average contact pressure between a foundation and the soil that can be sustained without triggering **shear failure** in the soil mass or causing **excessive settlement**.

It is categorized into:

- **Ultimate Bearing Capacity (q_u):** The theoretical maximum intensity of loading that causes the soil to fail in shear.
- **Net Ultimate Bearing Capacity (q_{nu}):** The ultimate capacity minus the initial overburden pressure ($q_{nu} = q_u - \gamma D_f$).
- **Safe Bearing Capacity (q_s):** The maximum pressure the soil can carry safely without shear failure, calculated by applying a **Factor of Safety (FoS)** (typically 2.5 to 3) to the net ultimate capacity.
- **Allowable Bearing Capacity (q_a):** The net loading intensity that ensures safety against *both* shear failure and excessive settlement. It is the lower of the safe bearing capacity and the pressure that limits settlement to permissible levels (e.g., 25mm).

Factors Affecting Bearing Capacity

- **Soil Properties:**
  - **Shear Strength Parameters:** Bearing capacity is a direct function of **cohesion (c)** and the **angle of internal friction (ϕ)**.
  - **Unit Weight (γ):** Higher soil density (unit weight) generally increases resistance and capacity.
- **Foundation Geometry:**

- o **Depth (D_f):** BC increases with depth due to the added weight (surcharge) of the surrounding soil, which provides confinement.
  - o **Width (B):** In cohesionless soils (sand), BC is proportional to the width; in purely cohesive soils (clay), BC is largely independent of width.
  - o **Shape:** Shape factors are applied for square, circular, or rectangular footings as they distribute stress differently than strip footings.
- **Groundwater Table:**
  - o The presence of water reduces BC by up to **50%** due to buoyancy, which reduces the effective unit weight of the soil.
- **Loading Conditions:**
  - o **Eccentricity:** Off-center loads reduce the effective area of the footing, significantly lowering the BC.
  - o **Inclination:** Loads applied at an angle reduce the vertical component of the capacity.
- **Physical Site Factors:**
  - o **Slopes:** Building on or near a slope introduces lateral forces that can destabilize the foundation.
  - o **Spacing:** Closely spaced footings can have overlapping stress zones, which may reduce individual BC if not accounted for.

Foundation Design Types

Foundations are selected based on the depth of the load-bearing strata and the magnitude of the structural load.

1. Shallow Foundations ($D_f \leq B$)

Typically used when the soil at shallow depths (usually < 3m) is sufficiently strong.

- **Isolated (Spread) Footing:** A square or rectangular pad supporting a single column. It is the most common and economical type.
- **Combined Footing:** Supports two or more columns when they are so close that individual footings would overlap.
- **Strip (Wall) Footing:** A continuous strip supporting a load-bearing wall or a row of very closely spaced columns.
- **Strap (Cantilever) Footing:** Two isolated footings connected by a rigid beam (strap) to manage eccentric loads, often near property lines.
- **Raft or Mat Foundation:** A large slab covering the entire building footprint. It is used for **weak soils** or heavy loads to distribute pressure and minimize **differential settlement**.

2. Deep Foundations ($D_f > B$)

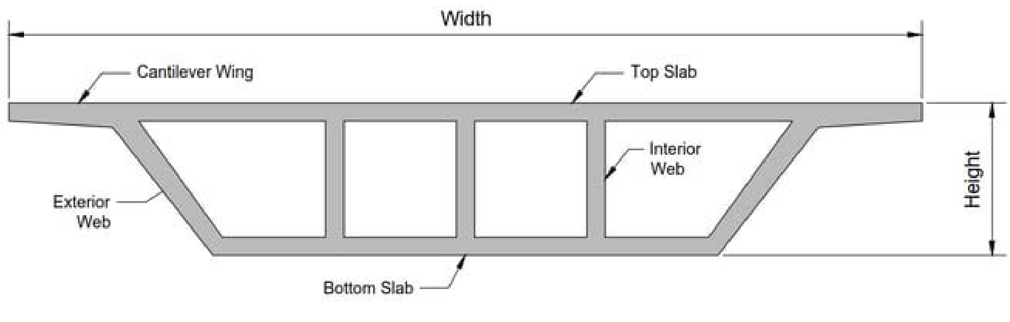
Used when hard strata are at great depths or when the structure faces heavy loads or uplift forces.

- **Pile Foundation:** Slender members (concrete, steel, or timber) driven or drilled into the ground. They transfer loads via **end-bearing** (to hard rock) or **skin friction** (against surrounding soil).
- **Well (Caisson) Foundation:** Large, hollow, watertight structures sunk into the ground. They are ideal for major **bridge piers** in rivers to resist heavy vertical and lateral forces (e.g., river flow, scouring).
- **Pier Foundation:** Large-diameter cylindrical columns (typically > 0.6m) used to transfer massive loads to hard strata, common in multi-story buildings and flyovers.

4. Describe the advantages of providing closed box shape in hollow box Girder Bridge and show the arrangement of closed box shape girder through a free hand diagram. [5]

Answer:

Hollow box girders consist of top and bottom slabs connected by vertical or inclined webs, forming a closed cell.



Advantages:

d) **High Torsional Rigidity:** The closed shape is extremely resistant to twisting, making it ideal for curved bridges.

e) **Structural Efficiency:** Offers high moment of inertia for less material weight, allowing for longer spans.

f) **Aesthetics:** Provides a clean, sleek appearance with a smooth bottom surface.
g) **Utility Space:** The hollow interior can be used to house utility pipes (water, telecommunications).
h) **Maintenance:** The interior is accessible for inspection of prestressing cables and concrete condition.

Arrangement:
A typical cross-section shows a top deck slab (carrying traffic), webs, and a bottom soffit slab.

Section-B (25 Marks)

5. A. In the Terai region on Nepal, there is heavy damages to the life and property due to frequent floods. Give major reasons to such damages and possible measures to address the issues in a sustainable manner. [5]
B. Where do we provide Guide Banks? Discuss on the design concept of Guide Banks. [5]

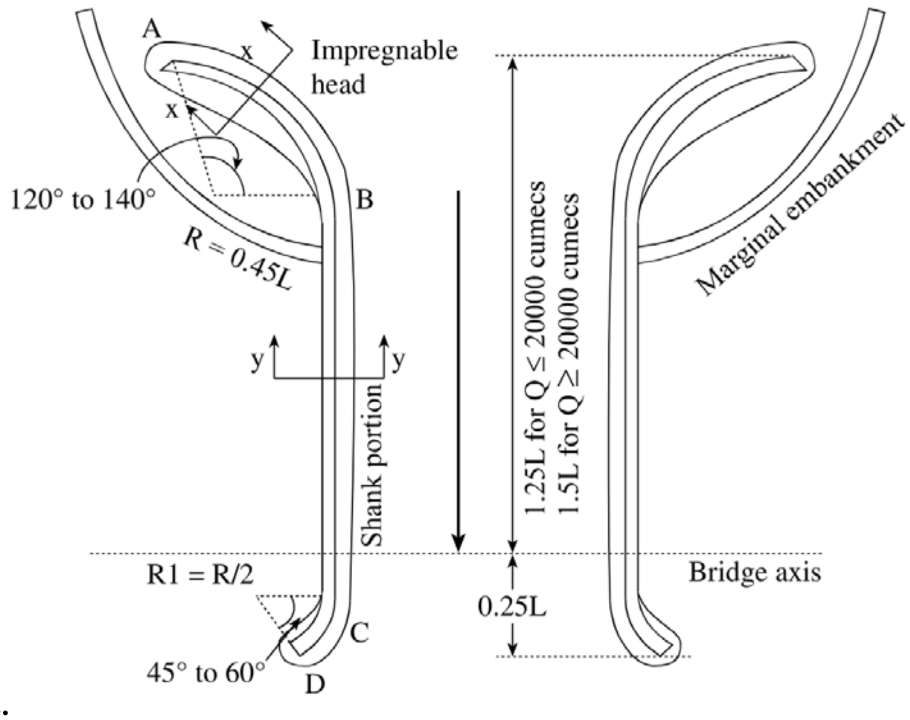
Answer (Part A):
Reasons for Terai Floods:

d) **Flat Topography:** Low longitudinal slope causes slow drainage of water.
e) **Deforestation in Chure:** Leads to high sediment loads that raise riverbeds (**aggradation**).
f) **Cross-border Obstructions:** Embankments and structures across the border can cause backwater effects.
g) **Extreme Rainfall:** Intense monsoon cloudbursts in the upstream hills.

Sustainable Measures:

d) **Watershed Management:** Reforestation in Chure and Mid-hills to reduce runoff and silt.
e) **Floodplain Zoning:** Restricting construction in flood-prone areas.
f) **Early Warning Systems (EWS):** Installing sensors to alert downstream communities.
g) **Bio-dykes:** Using bamboo and local vegetation to strengthen embankments.

Answer (Part B):
Guide Banks are provided to confine a meandering river into a narrow, fixed channel to pass through a bridge or barrage safely.



Design Concept:

c) Length: Usually $1L$ to $1.25L$ (where L is the length of the bridge).
d) Curvature: The upstream head is curved (bell-mouth) to ensure smooth entry of flow.
e) Launch Apron: A thick layer of stones/crates at the toe that "launches" (drops) into the scour hole during floods to protect the bank.
f) Slope Protection: Usually 1:2 or 1:3 slope protected by stone pitching or gabions.

6. What do you mean by Duty of Water? Explain the influence of several factors which affects duty and how duty of water can be improved. [10]

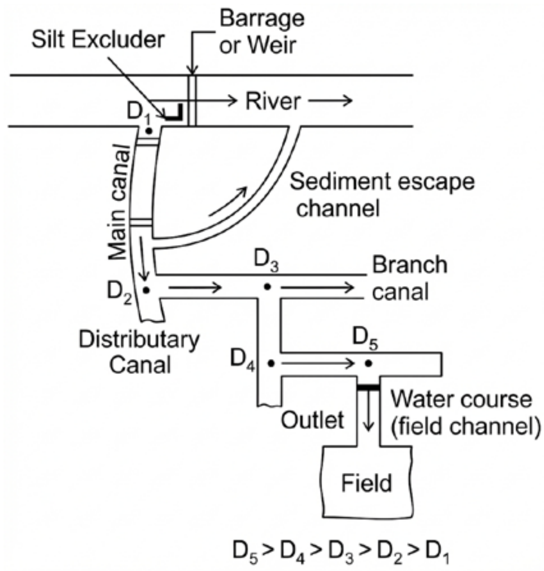
Answer:
Duty (D): The area of land (in hectares) that can be irrigated by a constant discharge of 1 cumec (m^3/s) throughout the base period of a crop. Unit: hectares/cumec.

$$\Delta = \frac{8.64B}{D}$$

Δ = Delta in cm

B= Base period or crop period of the crop

D = Duty of water in hectares/cumec



Factors Affecting Duty:

a) **Type of Crop:** Rice requires more water (Low Duty) than Maize (High Duty).
b) **Climate:** High temperature/wind increases evaporation, reducing duty.
c) **Soil Type:** Sandy soil has high percolation losses (Low Duty); clayey soil has better retention (High Duty).
d) **Method of Irrigation:** Drip/Sprinkler has higher duty than flood irrigation.
e) **Canal Lining:** Lined canals reduce seepage losses, increasing duty.
f) **Rainfall:** Effective rainfall reduces the need for irrigation, increasing duty.

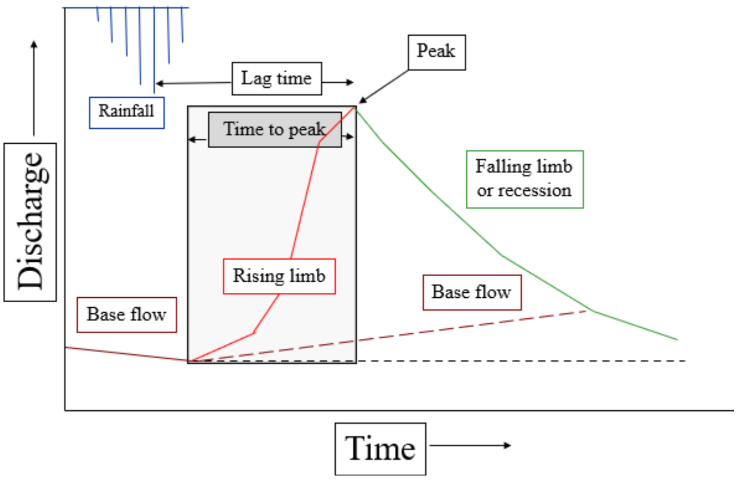
Methods to Improve Duty:

a) **Canal Lining:** To minimize seepage.
b) **Efficient Distribution:** Using gated outlets and rotating water supply (**Warabandi**).
c) **Land Leveling:** Ensures uniform water spread and less wastage.
d) **Volumetric Measurement:** Charging farmers based on volume encourages water saving.
e) **Proper Tillage:** Deep plowing improves soil moisture retention.

7. What is a hydrograph? Explain the use of hydrograph in rainfall assessment? [5]

Answer:

A Hydrograph is a graphical representation of the discharge (Q) of a stream or river as a function of time (t) at a specific location.



Use in Rainfall Assessment:

a) **Determining Peak Discharge:** Crucial for designing hydraulic structures like bridges and culverts.
b) **Estimating Runoff Volume:** The area under the hydrograph (minus baseflow) gives the total volume of direct runoff from a storm.
c) **Lag Time Calculation:** Helps determine the time delay between peak rainfall and peak flow.
d) **Developing Unit Hydrographs:** Used to predict the runoff response for any given rainfall duration/intensity.
e) **Flood Forecasting:** Analyzing how a catchment responds to monsoon rain to warn downstream areas.

Section-C (25 Marks)

8. Compare the rigid and flexible road pavement from various criteria. [10]

Answer:

Criteria	Flexible Pavement (Bituminous)	Rigid Pavement (RCC)
Load Transfer	Grain-to-grain contact (Layer by layer)	Slab action (Flexural strength)
Materials	Bitumen, Aggregates	Cement Concrete, Steel
Design Life	10–15 years	30–40 years
Initial Cost	Low	High
Maintenance	High (frequent repairs/overlays)	Low (only joint sealing)
Stage Construction	Possible	Not possible
Temperature	Softens in high heat	Thermal expansion/contraction
Visibility	Poor (Dark surface)	Good (Light surface)

9. What are the desirable properties of bituminous mixes? Briefly explain the ductility test of bitumen and its engineering application. [10]

Answer:

Desirable Properties of Bituminous Mixes:A well-designed mix balances several interrelated properties:

- **Stability:** The resistance to permanent deformation (e.g., shoving or grooving) under traffic loads.
- **Durability:** The ability to resist weathering, aging (oxidation), and abrasive actions from wheel loads.
- **Flexibility:** The capacity of the pavement to withstand bending and deflections under repeated traffic without developing cracks.
- **Workability:** The ease with which a mix can be laid, spread, and compacted at the construction site.
- **Skid Resistance:** The resistance provided against the skidding of vehicle tires, influenced by surface texture and bitumen content.
- **Impermeability:** Resistance to the entry of air and water into the mix, which prevents stripping and oxidative hardening.

Ductility Test:

The ductility test measures the distance in centimeters to which a standard bitumen sample can be stretched before it breaks.

Procedure:

a) **Preparation:** Melted bitumen is poured into a standard briquette mould (1 cm² cross-section at its narrowest) and allowed to cool.
b) **Conditioning:** The assembly is placed in a water bath at a standard temperature of **27°C** (or 25°C per some standards) for approximately 90 minutes.
c) **Testing:** The briquette clips are attached to a testing machine and pulled apart horizontally at a constant speed of **5 cm/min** (50 mm/min) until the thread of bitumen ruptures.
d) **Measurement:** The total distance pulled at the moment of rupture is recorded as the ductility value.

Engineering Application

- **Suitability Assessment:** It helps engineers determine if a bitumen binder is suitable for road construction. A minimum ductility value of **50 cm to 100 cm** is typically required for high-quality paving jobs.
- **Crack Prevention:** High ductility ensures the binder can form a thin film around aggregates that can accommodate expansion and contraction due to temperature changes, especially preventing brittle cracking in **cold climates**.
- **Adhesion & Cohesion:** The test reflects the adhesive and cohesive strength of the binder, which is critical for holding the pavement structure together under repeated heavy traffic.

10. What are the factors to consider in site selection of Airport? [5]

Answer:

a) **Wind Direction (Wind Rose):** Runways must be aligned with the prevailing wind direction for safe takeoff/landing.
b) **Topography:** Flat land is preferred to minimize earthwork (critical in Nepal's hills).
c) **Visibility:** Absence of frequent fog, smoke, or haze.
d) **Obstructions:** Area must be clear of tall mountains, buildings, or towers in the approach/take-off zones.
e) **Soil Conditions:** Good subgrade strength and natural drainage to support heavy aircraft loads.
f) **Future Expansion:** Availability of land for adding terminals or runways.

Section-D (25 Marks)

11. **What are the merits and demerits of slow sand filter and rapid gravity filter? Please compare on the basis of suitability, design and operation. [10]**

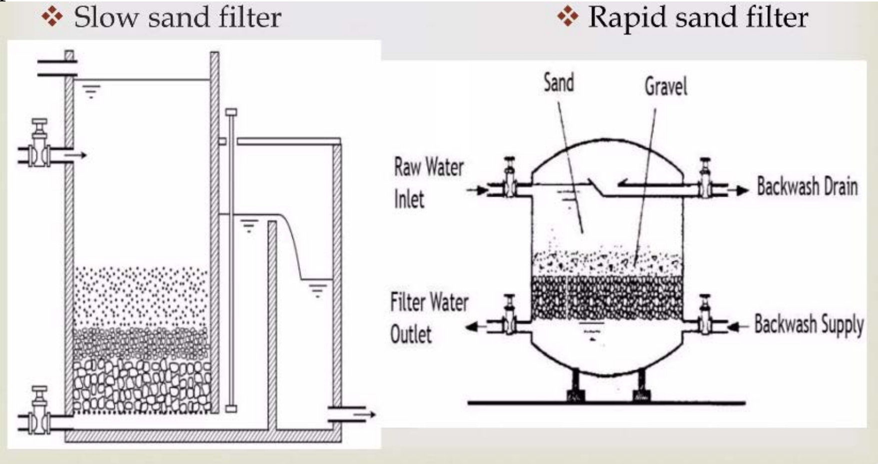
Answer:

Comparison of Merits and Demerits:

Feature	Slow Sand Filter (SSF)	Rapid Gravity Filter (RSF)

Merits	Simple to construct/operate; very high bacterial removal (99%+); requires minimal chemicals.	Compact (requires ~1/30th the area of SSF); high output rate; handles high turbidity.
Demerits	Requires massive land area; low filtration rate; less effective at removing color.	High operational cost (chemicals/energy); needs skilled labor; lower bacterial removal (80-90%).

Slow sand filters (SSF) and rapid gravity filters (RSF) differ significantly in their purification mechanisms, throughput, and operational demands.



Feature	Slow Sand Filter (SSF)	Rapid Gravity Filter (RSF)
	1. Suitability	
Ideal Location	Rural areas or small communities.	Large urban cities with limited land.
Water Source	Clearer raw water (turbidity < 30–50 NTU).	Highly turbid water (pre-treated).
Flexibility	Poor; cannot handle sudden demand spikes.	High; easily adaptable to demand changes.
	2. Design	
Filtration Rate	Slow: 100 – 200 L/h/m².	Fast: 3,000 – 6,000 L/h/m².
Sand Media	Finer: 0.25 – 0.35 mm (Effective Size).	Coarser: 0.45 – 0.70 mm (Effective Size).
Under-drainage	Simple: Open-jointed pipes or split tiles.	Complex: Manifold and laterals with nozzles.
Unit Area	Large units: 50 – 2,000 m².	Small units: 10 – 100 m².
	3. Operation	
Pre-treatment	Simple sedimentation (coagulation not needed).	Mandatory coagulation and flocculation.
Cleaning Method	Scraping the top 1.5–3 cm (Schmutzdecke).	Backwashing (upward flow of air and water).
Cleaning Interval	Long: 1 to 3 months.	Short: 24 to 72 hours.
Labor Skill	Low; minimal technical skill required.	High; requires skilled plant

12. **Describe the process of initial environmental examination (IEE) of a municipal solid waste management project. [10]**

Answer:

An **Initial Environmental Examination (IEE)** is an analytical study conducted to determine whether a proposed project will have significant adverse impacts on the environment and to identify measures to mitigate or minimize such effects. For municipal solid waste management, the process ensures that waste disposal and processing do not harm public health or local ecosystems.

The process is governed by the **Environment Protection Act (EPA), 2076** and the **Environment Protection Rules (EPR), 2077**.

A. Determining the Requirement (Project Thresholds)

Under **Rule 3** and **Schedule 2** of the EPR, 2077, a municipal solid waste management project requires an IEE if it meets any of the following technical criteria:

- **Landfilling:** Filling waste in land at a rate of **1,000 to 5,000 tons per year**.
- **Processing/Recovery:** Waste transfer stations, resource recovery areas, or composting plants covering an area of **5 to 10 hectares**.
- **Chemical/Biological Treatment:** Sorting or processing waste (chemical, mechanical, or biological) covering **5 to 10 hectares**.

B. The Step-by-Step IEE Process

The procedural journey of an IEE is structured as follows:

Step	Action	Legal Provision
Step 1: Preparation of ToR	The proponent must prepare a **Terms of Reference (ToR)** following the format in Schedule 7.	EPR Rule 5(1)
Step 2: ToR Approval	For local projects, the ToR is approved by the body designated by local law; for national projects, it is the concerned Federal Ministry.	EPA Sec 5, EPR Rule 5(2)
Step 3: Public Notice	A **7-day notice** must be published in a national daily newspaper to collect written suggestions from local stakeholders (schools, hospitals, etc.).	EPR Rule 4(2)
Step 4: Public Hearing	The proponent must conduct a **public hearing** in the project area to gather feedback from the affected community.	EPA Sec 3(5), EPR Rule 6
Step 5: Analysis of Alternatives	The study must include a detailed analysis of project **alternatives** (location, design, technology) and provide justifications for the chosen path.	EPA Sec 4(1)
Step 6: Field Study & Drafting	Experts conduct a study to identify biological, physical, and socio-economic impacts as per the approved ToR.	EPR Rule 7
Step 7: Report Submission	The IEE report, following the format in **Schedule 11**, must be submitted for approval within **2 years** of ToR approval.	EPR Rule 7(5), Rule 8(11)
Step 8: Review & Approval	The concerned body reviews the report and approves it within **15 days** if no significant environmental threats are identified.	EPA Sec 7(5), EPR Rule 9(8)